



Fig. 2: Author's prototype

$100 \times I_m \times R$ (where I_m is the current to be measured). The range can be controlled by selecting R (1 k Ω , 10 k Ω and 100 k Ω) appropriately.

The diode in parallel with the meter protects the circuit against

high currents that may occur due to improper range selection. Further, the range resistors do not have any noticeable effect on the current to be measured. For instance, to produce a current of 10 nA from a 5V source, it is necessary to include a 500-mega-ohm

resistor in the circuit. Since a 100k Ω is being used to sample this current, it is very small as compared to the large value 500M Ω .

Any meter can be used to calibrate the output, provided the gain of the

DC amplifier is changed appropriately. That is, a meter with full-scale deflection of 150 μ A would require a gain of 150 and vice-versa. Using a commonly available digital multimeter, adjust the calibrator (preset VR2) such that the resistance of the calibrator and 100 μ A meter (typically, which has a resistance of 700 ohms) in series is exactly 1 kilo-ohm. Now wire the entire circuit and adjust the set-zero trimpot to get zero deflection in any range of the multimeter. Now the nano-ampere meter is ready for use.

Assemble the circuit on a general-purpose PCB and enclose in a small cabinet. Fix the ammeter on the front panel. Provide three terminals for the power supply (-5V, +5V and GND) on the rear panel and two terminals on the front panel for the current to be measured. ●

CURTAIN OPENER

■ SUNIL KUMAR

Open or close the curtains of your house at the press of a switch. All you need is this

electronic curtain opener. Meant for domestic use, especially for small doors and windows, this curtain opener is basically a geared motor along with mechanical assembly (in-

cluding the reed switches fitted to the door or window) to open and close the curtain.

